

Fundamentals of Electrical Engineering: Mesh Current Analysis

High-Level Overview

Mesh Current Analysis (also referred to as the Mesh Current Method) is a structured, algorithmic technique used to analyze planar electrical circuits. It allows engineers to determine all unknown currents and voltages across any circuit component systematically.

Fundamental Terminology

- **Loop:** Any closed path in a circuit that starts at a specific node, passes through a set of circuit components, and returns to the starting node without crossing any node more than once.
- **Mesh:** A special type of loop that **does not contain any other loops inside it**. Think of a window pane: the whole window frame is a loop, but each individual pane of glass inside it is a "mesh". Mesh Analysis only applies to **planar circuits** (circuits that can be drawn on a 2D plane without any wires crossing over each other).
- **Mesh Current:** An imaginary, fictitious current assumed to circulate continuously around the perimeter of a specific mesh. While branch currents represent the physical movement of charge through a specific component, mesh currents are mathematical conveniences used to write simple linear equations.
- **Kirchhoff's Voltage Law (KVL):** The foundational principle underlying Mesh Analysis. KVL states that the algebraic sum of all electric potential differences (voltages) around any closed loop in a circuit must equal zero ($\sum V = 0$). This is a direct consequence of the **Law of Conservation of Energy**.
- **Ohm's Law:** Relates potential difference (V), current (I), and resistance (R) across a component:

$$V = I \cdot R$$

Core Theoretical Principles

1. Active vs. Passive Elements

- **Active Elements:** Components that supply energy to the circuit (e.g., independent voltage sources, independent current sources).
- **Passive Elements:** Components that consume or store energy (e.g., resistors, capacitors, inductors).

2. Source Transformation Rule

Before applying the standardized KVL formulation, any active **current source** connected in parallel with a resistor should ideally be converted into an equivalent **voltage source** in series with that same resistor.

$$\text{Equivalent Voltage Source: } V_s = I_s \cdot R_s$$

- **Voltage Source Conversion:** The positive terminal of the converted voltage source corresponds to the direction the current arrow was pointing.
- **Resistor Placement:** The parallel resistor R_s is moved into a series connection with the new voltage source.

The Standard 3-Step Algorithm for Mesh Analysis

To solve any planar circuit using Mesh Analysis, follow these systematic steps:

Step 1: Convert Active Elements

Convert all active current sources (if present) into equivalent voltage sources using Source Transformation. Ensure that all independent sources in the circuit are expressed as voltage sources.

Step 2: Define and Mark Mesh Currents

Identify all fundamental meshes (window panes) in the planar circuit. Assign an imaginary loop current to each mesh ($i_1, i_2, i_3, \dots, i_n$).

- **Convention:** Always assign all mesh currents in the **clockwise direction**. This standardizes the sign convention and eliminates sign errors during equation setup.

Step 3: Apply KVL and Formulate Mesh Equations

Write a KVL equation for each mesh by traversing clockwise around the loop:

1. **Self-Impedance Voltage Drop:** The current passing through the mesh's own total resistance produces a drop: $-(R_{\text{total, mesh}}) \cdot i_{\text{mesh}}$.
2. **Mutual/Shared Resistor Voltage Contribution:** If a resistor is shared between the current mesh i_k and a neighboring mesh i_m , the neighboring mesh current flows in the opposite direction through that resistor, producing a positive voltage contribution: $+(R_{\text{shared}}) \cdot i_m$.
3. **Voltage Sources:** Traversing a voltage source from its negative to positive terminal represents a voltage **rise** ($+V$). Traversing from positive to negative represents a voltage **drop** ($-V$).

Matrix Form of General Mesh Equations

For an n -mesh circuit, the system of linear equations takes the standard matrix form:

$$\begin{bmatrix} R_{11} & R_{12} & \dots & R_{1n} \\ R_{21} & R_{22} & \dots & R_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ R_{n1} & R_{n2} & \dots & R_{nn} \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ \vdots \\ i_n \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

- R_{kk} is the sum of all resistances in mesh k .
- R_{km} ($k \neq m$) is the mutual resistance shared between mesh k and mesh m (typically negative or accounted for via algebraic sign rules).
- v_k is the net sum of driving voltage sources in mesh k .

Symbolic Derivation & Analysis of the Wheatstone Bridge

Consider a classic Wheatstone Bridge circuit configuration connected to a DC supply V_S with internal resistance R_S , containing a central Galvanometer with internal resistance R_5 .

Diagram: Wheatstone Bridge Circuit with Mesh Currents

Visual Description of Circuit:

The circuit forms a diamond-shaped bridge network between two central vertical nodes marked **A** (top) and **B** (bottom).

- **Top-Left Arm:** Resistor R_1 connects the left central node to Node **A**.
- **Bottom-Left Arm:** Resistor R_2 connects the left central node to Node **B**.
- **Top-Right Arm:** Resistor R_3 connects Node **A** to the right central node.
- **Bottom-Right Arm:** Resistor R_4 connects Node **B** to the right central node.
- **Central Vertical Arm:** A Galvanometer denoted by a circled letter **G** with internal resistance R_5 connects Node **A** directly down to Node **B**.
- **External Loop:** Below the bridge, an external loop connects the left central node to the right central node, consisting of a series resistor R_S and a DC voltage source V_S (positive terminal on the right, negative on the left).
- **Mesh Current Assignments:**
 - Mesh 1 (i_1): Clockwise loop formed inside the left triangle (R_1, R_5, R_2).
 - Mesh 2 (i_2): Clockwise loop formed inside the right triangle (R_3, R_4, R_5).
 - Mesh 3 (i_3): Clockwise loop formed in the bottom loop containing R_2, R_4 , source resistor R_S , and voltage source V_S .

Step-by-Step Derivation of KVL Mesh Equations

Mesh 1 Analysis (Left Loop with Current i_1)

Traversing clockwise through Mesh 1:

- Drop across R_1 : $-R_1 i_1$

- Drop across Galvanometer branch R_5 : i_1 flows down through R_5 , while i_2 flows up through R_5 . Net current downwards is $(i_1 - i_2)$. Drop is $-R_5(i_1 - i_2)$.
- Drop across R_2 : i_1 flows down-left through R_2 , while i_3 flows down-right through R_2 . Net current is $(i_1 - i_3)$. Drop is $-R_2(i_1 - i_3)$.

$$\sum V = -R_1 i_1 - R_5(i_1 - i_2) - R_2(i_1 - i_3) = 0$$

Expanding terms:

$$-R_1 i_1 - R_5 i_1 + R_5 i_2 - R_2 i_1 + R_2 i_3 = 0$$

Grouping terms by mesh currents i_1 , i_2 , and i_3 :

$$-(R_1 + R_2 + R_5)i_1 + (R_5)i_2 + (R_2)i_3 = 0 \quad \text{--- (Equation 1)}$$

Mesh 2 Analysis (Right Loop with Current i_2)

Traversing clockwise through Mesh 2:

- Drop across R_3 : $-R_3 i_2$
- Drop across R_4 : i_2 flows down-left through R_4 , while i_3 flows down-right. Net current through R_4 is $(i_2 - i_3)$. Drop is $-R_4(i_2 - i_3)$.
- Drop across R_5 : i_2 flows up through R_5 , while i_1 flows down. Net current upwards is $(i_2 - i_1)$. Drop is $-R_5(i_2 - i_1)$.

$$\sum V = -R_3 i_2 - R_4(i_2 - i_3) - R_5(i_2 - i_1) = 0$$

Expanding and grouping terms:

$$(R_5)i_1 - (R_3 + R_4 + R_5)i_2 + (R_4)i_3 = 0 \quad \text{--- (Equation 2)}$$

Mesh 3 Analysis (Bottom Outer Loop with Current i_3)

Traversing clockwise through Mesh 3:

- Drop across R_2 : i_3 opposes i_1 . Net drop is $-R_2(i_3 - i_1)$.
- Drop across R_4 : i_3 opposes i_2 . Net drop is $-R_4(i_3 - i_2)$.
- Rise/Drop across V_S : Entering negative, exiting positive $\rightarrow +V_S$.
- Drop across R_S : $-R_S i_3$.

$$\sum V = -R_2(i_3 - i_1) - R_4(i_3 - i_2) + V_S - R_S i_3 = 0$$

Expanding and grouping terms:

$$(R_2)i_1 + (R_4)i_2 - (R_2 + R_4 + R_S)i_3 = -V_S \quad \text{--- (Equation 3)}$$

Fully Worked Numerical Examples

Illustration 1: Galvanometer Current Determination

Problem Statement:

For the bridge circuit shown below, determine the exact current passing through the galvanometer (I_{AB}) from Node **A** to Node **B**.

 Diagram: Illustration 1 Circuit Diagram

Visual Description of Circuit:

Diamond bridge configuration with node **A** at top and node **B** at bottom:

- $R_1 = 40 \, \Omega$ (top-left)
- $R_2 = 20 \, \Omega$ (bottom-left)
- $R_3 = 30 \, \Omega$ (top-right)
- $R_4 = 30 \, \Omega$ (bottom-right)
- Galvanometer resistance between A and B: $R_5 = 100 \, \Omega$
- External path: Series resistor $R_S = 10 \, \Omega$ and DC source $V_S = 10 \, \text{V}$.

Step-by-Step Solution

Step 1: Assign Mesh Currents

Define three clockwise mesh currents:

- i_1 : Top-left mesh
- i_2 : Top-right mesh
- i_3 : Bottom loop mesh

Step 2: Formulate KVL Linear Equations

For Mesh 1:

$$-(R_1 + R_2 + R_5)i_1 + (R_5)i_2 + (R_2)i_3 = 0$$

$$-(40 + 20 + 100)i_1 + 100i_2 + 20i_3 = 0$$

$$-160i_1 + 100i_2 + 20i_3 = 0$$

Divide by 20:

$$-8i_1 + 5i_2 + i_3 = 0 \implies i_3 = 8i_1 - 5i_2 \quad \text{--- (Eq. A)}$$

For Mesh 2:

$$(R_5)i_1 - (R_3 + R_4 + R_5)i_2 + (R_4)i_3 = 0$$

$$100i_1 - (30 + 30 + 100)i_2 + 30i_3 = 0$$

$$100i_1 - 160i_2 + 30i_3 = 0$$

Divide by 10:

$$10i_1 - 16i_2 + 3i_3 = 0 \quad \text{--- (Eq. B)}$$

For Mesh 3:

$$(R_2)i_1 + (R_4)i_2 - (R_2 + R_4 + R_5)i_3 = -V_S$$

$$20i_1 + 30i_2 - (20 + 30 + 10)i_3 = -10$$

$$20i_1 + 30i_2 - 60i_3 = -10$$

Divide by 10:

$$2i_1 + 3i_2 - 6i_3 = -1 \quad \text{--- (Eq. C)}$$

Step 3: Solve System of Equations

Substitute **Eq. A** ($i_3 = 8i_1 - 5i_2$) into **Eq. B**:

$$10i_1 - 16i_2 + 3(8i_1 - 5i_2) = 0$$

$$10i_1 - 16i_2 + 24i_1 - 15i_2 = 0$$

$$34i_1 - 31i_2 = 0 \implies i_2 = \frac{34}{31}i_1 \quad \text{--- (Eq. D)}$$

Express i_3 in terms of i_1 using **Eq. D**:

$$i_3 = 8i_1 - 5 \left(\frac{34}{31}i_1 \right) = \left(8 - \frac{170}{31} \right) i_1 = \left(\frac{248 - 170}{31} \right) i_1 = \frac{78}{31}i_1 \quad \text{--- (Eq. E)}$$

Substitute **Eq. D** and **Eq. E** into **Eq. C**:

$$2i_1 + 3 \left(\frac{34}{31}i_1 \right) - 6 \left(\frac{78}{31}i_1 \right) = -1$$

$$2i_1 + \frac{102}{31}i_1 - \frac{468}{31}i_1 = -1$$

Multiply entire equation by 31:

$$62i_1 + 102i_1 - 468i_1 = -31$$

$$-304i_1 = -31$$

$$i_1 = \frac{31}{304} \text{ A} \approx 0.10197 \text{ A}$$

Now find i_2 :

$$i_2 = \frac{34}{31} \left(\frac{31}{304} \right) = \frac{34}{304} \text{ A} \approx 0.11184 \text{ A}$$

Step 4: Calculate Target Branch Current I_{AB}

The branch connecting Node **A** to Node **B** contains the Galvanometer.

- Current i_1 flows downwards from **A** to **B**.
- Current i_2 flows upwards from **B** to **A**.

The net current flowing from **A** to **B** is:

$$I_{AB} = i_1 - i_2$$

$$I_{AB} = \frac{31}{304} - \frac{34}{304} = -\frac{3}{304} \text{ A} \approx -0.00987 \text{ A} = -9.87 \text{ mA}$$

Physical Interpretation: The negative sign simply indicates that the physical current actually flows in the direction opposite to our assumed direction —meaning 9.87 mA flows upwards from Node **B** to Node **A**.

Illustration 2: Mixed Source Circuit Analysis

Problem Statement:

For the multi-loop circuit given below:

1. Determine the exact current passing through the 2Ω resistor.
2. Determine the electric potential difference between Node **A** and Node **B** ($V_A - V_B$).

Visual Description of Circuit:

- **Left Branch:** A 5 V independent DC voltage source (positive terminal pointing up) with node **A** located at its top terminal.
- **Top Outer Branch:** A $16\ \Omega$ resistor connected horizontally between node **A** and node **B**.
- **Middle Horizontal Line:** Resistor $R_{8\Omega} = 8\ \Omega$ connected between node **A** and a middle node. Resistor $R_{12\Omega} = 12\ \Omega$ connected between the middle node and node **B**.
- **Vertical Resistor:** Resistor $R_{2\Omega} = 2\ \Omega$ connected vertically between the middle node and the bottom common reference rail.
- **Right Branch:** A parallel branch consisting of an independent current source of 0.5 A (arrow pointing upwards) connected to node **B**, and a $4\ \Omega$ resistor connected between node **B** and the bottom rail.

Step-by-Step Solution

Step 1: Perform Source Transformation

Notice the rightmost active branch: a current source of 0.5 A in parallel with a $4\ \Omega$ resistor connected to Node **B**.

Convert this current source into an equivalent voltage source:

$$V_{eq} = I \cdot R = 0.5\text{ A} \times 4\ \Omega = 2\text{ V}$$

- The direction of the arrow points upward towards Node **B**, so the positive terminal of V_{eq} points upward toward Node **B**.
- The $4\ \Omega$ resistor is placed in series with this 2 V source.

Step 2: Define Mesh Currents

After transformation, the circuit consists of three clean meshes:

- **Mesh 1 (Top Loop):** Current i_1 circulating clockwise through $16\ \Omega$, $12\ \Omega$, and $8\ \Omega$.
- **Mesh 2 (Bottom-Left Loop):** Current i_2 circulating clockwise through $8\ \Omega$, $2\ \Omega$, and 5 V source.
- **Mesh 3 (Bottom-Right Loop):** Current i_3 circulating clockwise through $12\ \Omega$, the transformed $4\ \Omega$ branch with 2 V source, and $2\ \Omega$.

Step 3: Set Up KVL Equations

Mesh 1:

$$-16i_1 - 12(i_1 - i_3) - 8(i_1 - i_2) = 0$$

$$-36i_1 + 8i_2 + 12i_3 = 0$$

Divide by 4:

$$-9i_1 + 2i_2 + 3i_3 = 0 \quad \text{--- (Eq. 1)}$$

Mesh 2:

$$5 - 8(i_2 - i_1) - 2(i_2 - i_3) = 0$$

$$8i_1 - 10i_2 + 2i_3 = -5 \quad \text{--- (Eq. 2)}$$

Mesh 3:

$$-12(i_3 - i_1) - 2 - 4i_3 - 2(i_3 - i_2) = 0$$

$$12i_1 + 2i_2 - 18i_3 = 2$$

Divide by 2:

$$6i_1 + i_2 - 9i_3 = 1 \quad \text{--- (Eq. 3)}$$

Step 4: Solve the Equations System

From **Eq. 3**, express i_2 :

$$i_2 = 1 - 6i_1 + 9i_3 \quad \text{--- (Eq. 4)}$$

Substitute **Eq. 4** into **Eq. 1**:

$$-9i_1 + 2(1 - 6i_1 + 9i_3) + 3i_3 = 0$$

$$-9i_1 + 2 - 12i_1 + 18i_3 + 3i_3 = 0$$

$$-21i_1 + 21i_3 = -2 \implies i_1 - i_3 = \frac{2}{21} \implies i_1 = i_3 + \frac{2}{21} \quad \text{--- (Eq. 5)}$$

Substitute **Eq. 4** and **Eq. 5** into **Eq. 2**:

$$8 \left(i_3 + \frac{2}{21} \right) - 10(1 - 6i_1 + 9i_3) + 2i_3 = -5$$

$$8i_3 + \frac{16}{21} - 10 + 60 \left(i_3 + \frac{2}{21} \right) - 90i_3 + 2i_3 = -5$$

$$8i_3 + \frac{16}{21} - 10 + 60i_3 + \frac{120}{21} - 90i_3 + 2i_3 = -5$$

Combine i_3 terms:

$$(8 + 60 - 90 + 2)i_3 + \frac{136}{21} - 10 = -5$$

$$-20i_3 + \frac{136}{21} - 5 = 0$$

$$-20i_3 = 5 - \frac{136}{21} = \frac{105 - 136}{21} = -\frac{31}{21}$$

$$i_3 = \frac{31}{420} \text{ A} \approx 0.0738 \text{ A}$$

Using **Eq. 5**, find i_1 :

$$i_1 = \frac{31}{420} + \frac{2}{21} = \frac{31 + 40}{420} = \frac{71}{420} \text{ A} \approx 0.1690 \text{ A}$$

Using **Eq. 4**, find i_2 :

$$i_2 = 1 - 6 \left(\frac{71}{420} \right) + 9 \left(\frac{31}{420} \right) = 1 - \frac{426}{420} + \frac{279}{420} = \frac{420 - 426 + 279}{420} = \frac{273}{420} \text{ A} = 0.65 \text{ A}$$

Step 5: Calculate Required Quantities

1. Current through 2Ω resistor ($I_{2\Omega}$):

The 2Ω resistor is shared between Mesh 2 and Mesh 3.

$$I_{2\Omega} = i_2 - i_3 = 0.65 \text{ A} - 0.0738 \text{ A} = 0.5762 \text{ A} \quad (\text{flowing downwards})$$

2. **Potential Difference** $V_{AB} = V_A - V_B$:

The $16\,\Omega$ resistor connects Node **A** directly to Node **B**. The current flowing through this branch from Node **A** to Node **B** is i_1 .
Applying Ohm's Law across the $16\,\Omega$ branch:

$$V_{AB} = i_1 \cdot R_{16\Omega} = \left(\frac{71}{420}\,\text{A}\right) \times 16\,\Omega = \frac{1136}{420}\,\text{V} \approx 2.705\,\text{V}$$

Comparative Summary & Key Takeaways

Feature / Aspect	Mesh Analysis Method	Nodal Analysis Method
Primary Variable	Fictitious Loop/Mesh Currents (i)	Node Voltages (v)
Governing Law	Kirchhoff's Voltage Law (KVL)	Kirchhoff's Current Law (KCL)
Applicability	Planar Circuits only	All Circuits (Planar & Non-planar)
Source Preference	Voltage Sources (Current sources transformed)	Current Sources (Voltage sources transformed)
Equation Count	Number of Meshes ($M = E - N + 1$)	Number of Non-Reference Nodes ($N - 1$)

Quick Problem-Solving Reference Checklist

- Verify Planarity:** Ensure wires do not cross over each other without connecting.
- Transform Current Sources:** Convert parallel (I_s, R_s) combinations to series (V_s, R_s) pairs.
- Assign Clockwise Currents:** Assign $i_1, i_2, \dots, i_n$ to each mesh consistent with clockwise flow.
- Formulate KVL Equations:** For each mesh k , calculate $(-\sum R)i_k + \sum (R_{\text{shared}}i_{\text{neighbor}}) + \sum V_{\text{sources}} = 0$.
- Solve System:** Use substitution, matrix inversion, or Cramer's rule to find all mesh currents.
- Determine Branch Variables:** Obtain actual physical branch currents via super-position of overlapping mesh currents ($I_{\text{branch}} = i_{\text{same}} - i_{\text{opposite}}$).